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Synesthesia: perspectives from cognitive neuroscience

Article in *Journal of Consciousness Studies* · January 2005

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Book Reviews

Lynn C. Robertson and Noam Sagiv (ed.)

Synesthesia: Perspectives from Cognitive Neuroscience

New York: Oxford University Press, 2005, 266 pp.

ISBN 0-19-516623-X

Reviewed by Richard E. Cytowic

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Twenty-five years ago, my colleagues warned me not to pursue synaesthesia because it was ‘too weird, too New Age’. ‘It will ruin your career,’ they prophesied, wide of the mark and not even wrong to judge by the renaissance of interest in a phenomenon often regarded as wholly subjective. Among the flurry of publications occurring since reflexive hostility has waned is this edited collection by **Lynn Robertson** and **Noam Sagiv**. Their book stems from a Cognitive Neuroscience Society symposium, with added chapters by two synaesthetes and others who have studied the phenomenon’s cognitive/neural basis or its development. The book ably reflects synaesthesia’s phenotypic diversity.

Synaesthesia bears on normal cognition: perception, attention, consciousness, memory, learning, language, thought and development. Evidence from infant behaviour, cortical plasticity, cross-modal matching, metaphor and language, as well as synaesthesia, suggests that connections — either direct or indirect — among brain regions typically associated with discrete modalities underlie each of these phenomena. Therefore elucidation of one topic can inform us about the others. While we learn much about normal cognition by studying deficits, there is much to learn from positive phenomena such as synaesthesia. A central focus of present synaesthesia research is on developing methods to confirm its perceptual reality. Most commonly, one finds a blend of behavioural data substantiating synaesthesia’s automatic perceptual nature, and functional imaging studies showing activation of, for example, colour areas during reported synaesthetic colour experience.

Sean Day reviews synaesthesia’s demographics and socio-cultural aspects, also addressing widespread misinformation, mysticism, and pseudoscience about it being a sixth sense or somehow paranormal. He especially addresses the

lifelong frustration subjects feel at not being believed. It is especially galling that doctors, psychology professors, and similarly exalted professionals react with disbelieving disdain when their comfortable view of how perception ‘should be’ is challenged by first-hand experience of how it actually is.

Blake et al. describe a series of experiments demonstrating synaesthesia’s automatic perceptual nature and also explore the degree to which attention and awareness affect the resulting experience. They describe speeded reaction time tasks, the use of ambiguous figures, and Stroop-like tasks using stimuli either congruent or incongruent with a given subject’s synaesthesia. For example, a visual search that relies on the pop out of synaesthetically coloured oddball stimuli affects reaction time. Through such means they show that synaesthetic colour can be available before conscious awareness of the identity of the inducing stimulus.

Smilek and Dixon discuss grapheme–colour synaesthetes who ‘project’ or experience colour as an overlap appearing atop — meaning visually bound to — printed graphemes. They present evidence that synaesthetic colours are bound to graphemes before subjects attend to and become aware of them. The meaning of graphemes plays a role in binding given that the colour of an ambiguous figure depends on how subjects interpret it. Furthermore, experimental evidence is consistent with subjective reports: synaesthetes claim to perceive coloured graphemes rather than black graphemes that subsequently become coloured. The implication is that graphemes are processed as meaningful whole objects.

Savig and Robertson ask what synaesthesia shares in common with normal perception. The answer is a lot. They review evidence showing synaesthetic correspondences to be prevalent in human cognition, then ask whether synaesthetic binding obeys rules of normal binding of surface properties and whether attention plays a central role as it appears to do in normal perception. They discuss synaesthesia as a positive phenomenon, a kind of hyper-binding, and suggest several possibilities for its neural substrate.

The late **Jeffrey Gray** recaps how synaesthesia is a window on the hard problem of consciousness, summarizing his many years of thought on the issue.

Ramachandran and Hubbard aspire to inspire new interest in synaesthesia so that it gets the attention it deserves. They take on the philosophical riddle of qualia, suggesting how the study of synaesthesia can provide hints about the evolution, functional significance, and neural correlates of these entities. They make much of the fact that the grapheme and colour V4 area lie adjacent to one another on the fusiform gyrus. Their studies with fusiform and angular gyrus synaesthetes indicate that synaesthetes are behaviourally heterogeneous. It is thus premature to perform group studies, and existing ones need cautious interpretation. They further lend support for a left-hemispheric lateralization of synaesthesia, which I first proposed in 1989, and end by speculating why synaesthesia is more common among artistic individuals.

The two best chapters are those by **Daphne Maurer and Mondlach**, reevaluating the former’s hypothesis of neonatal synaesthesia, and **Lawrence Marks**, weighing the constraints imposed by development on competing theories of

synaesthesia. The latter is a tour de force. Specifically **Marks and Odgaard** lay out how neural theories thus far proposed fail to account for the learning that must underlie the development of any kind of synaesthesia involving culturally specific artifacts such as integers and letters (such lexical synaesthesia accounts for two-thirds of all instances). Each of the existing theories suffers from limits in its ability to explain all varieties of synaesthesia observed.

Such limitations point to a possibility I raised two decades ago, namely that synaesthesia may not be a singular entity. Nonetheless, synaesthesia has moved from an era of vague phenomenology to one of experimental research and shows itself to be far more than a strange curiosity.

The book's one shortcoming is having no chapter on genetics. Synaesthesia is an ideal condition to explore perceptual genomics, the study of how genes affect the way we perceive the world. Since it clusters in families, is not terribly uncommon, and can with care be confidently phenotyped, it provides a condition that can be traced through pedigrees for family linkage analysis.

Ran R. Hassin, James S. Uleman and John A. Bargh (ed.)

The New Unconscious

Oxford University Press, 2005, 604 pp., ISBN 0-19-514995-5 (hbk)

Reviewed by Justus de Swart

Does consciousness provide an exception to the rule that humans are basically automatons? By bringing together recent research in the fields of social cognition and social neuroscience, this book seems to suggest that there may be no exceptions. As will be described later, though, human 'automatons' may not necessarily be 'mechanistic'. One of the editors (Uleman) asks: 'What, if anything, cannot be done without awareness? What is consciousness for?' (p.6). Instead of tackling these questions head-on, a kind of 'subtraction' approach is adopted. The procedure is to exclude, one-by-one, mental phenomena having an unconscious basis, to see if any conscious residue remains. Interestingly, though it complicates matters, current research into the new unconscious shows that conscious and unconscious processes share some characteristics.

The book is divided into sections, starting off with 'Fundamental Questions', which takes a refreshing look at the familiar sticky problems over homunculi and free will. Next comes 'Basic Mechanisms', to do with what is known about the unconscious, most of it being empirically based. The last three sections explore the importance of the new unconscious to aspects of (social) mentality commonly thought to belong exclusively to consciousness. These are titled 'Intention', 'Perceiving and Engaging Others' and 'Self-Regulation'.

There is an impressive amount of experimental data supporting the view that a great part of human mentality - from perception, similarity detection and counterfactual thinking to unconscious metacognition (planning), goal-pursuit and working-memory - is reined in by the 'four horsemen of automaticity' (p.